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Flightworthiness Verification of a 3U CubeSat for Sounding Rocket Payloads with Ansys FEA

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17 January 2021

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Dear Professor Owen,

This report, entitled "Flightworthiness Verification of a 3U CubeSat for Sounding Rocket Payloads with Ansys FEA", was written to fulfill my WKRPT 200 requirements. This is my first work report submission.

The purpose of this report is to document the flightworthiness analysis of the Waterloo Rocketry's 2021 payload subsystem. The 2021 payload includes an experiment comparing the radiation shielding of boron nitride nanotube-thermoplastic polyurethane and metal oxide-acrylic composites. The experiment is fitted within a 3U CubeSat that is designed to be flown in Kraken of the Sky, the next hybrid engine sounding rocket being developed by Waterloo Rocketry.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I would like to thank Kyle Tam, Roman Kobets, Nicholas Christopher, and other Waterloo Rocketry team members for providing their advice and experience on this project. I received no other assistance.

Best regards,

Zhen Bo Bian
3A Mechanical Engineering

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Summary

This report details the analysis of a 3U CubeSat's flightworthiness for a sounding rocket flight to 30 000 feet. Analysis specifications are derived from previous flight data and from simulation data. Ansys FEA is used to evaluate the CubeSat's performance in the test environment. From preliminary results, the CubeSat is flightworthy and no changes are made to the structure.

Static structural, modal, and random vibration analysis are pursued to evaluate flightworthiness. Static structural analysis is used to evaluate the CubeSat under inertial loads due to rocket acceleration. Modal analysis is used to evaluate the CubeSat's dynamic behaviour. Random vibration analysis is used to evaluate the CubeSat under ASD loads due to the launch vehicle environment.

Inertial load test cases are derived from acceleration data from accelerometers and simulations. -Z direction loading is set to 15G. +Z direction loading is set to 5G. X and Y direction loading is set to 12G. In all completed test cases, maximum von Mises stress does not exceed 70 MPa.

The first mode of the CubeSat is calculated to be 718 Hz.

ASD loads are derived from acceleration data from a past rocket flight. The ASDs are derived to meet NASA-STD-7001B protoqualification vibration test specifications. enDAQ's VibrationData Toolbox software is used to generate ASDs. Random vibration analysis is not completed.

The CubeSat is modelled with tetrahedral elements. The internal modules are approximated with point masses. Body-body beam contacts are used to model screws.

1.0 Introduction

Waterloo Rocketry is a student design team of the University of Waterloo that is focused on rocket development. It draws undergraduate students from numerous departments to design, fabricate, and launch rockets [1]. Rockets are designed to be flown in the Intercollegiate Rocket Engineering Competition (IREC) of the Spaceport America Cup (SA Cup) and the Advanced Launch Challenge of the Launch Canada Challenge. With a target altitude of 30 000 feet, these competitions provide an unparalleled opportunity for students to learn, practice, and apply engineering skills towards developing a sounding rocket.

The rocket is divided into multiple components, one of which is the payload. The payload is an experiment or mechanical device that is flown with the rocket. The payload is entered in the Space Dynamics Laboratory (SDL) Payload Challenge of the SA Cup, a challenge intended to promote students to develop and fly experiments in their rockets [2].

1.1 CubeSat

CubeSats are a standardized type of picosatellite developed to increase accessibility of deploying small payloads to space [3]. Since its inception in 1999, the CubeSat Program has helped deploy numerous payloads by universities, high schools, and private firms due to the reduced cost and development time and increased accessibility to launch opportunities [3].

The CubeSat Program has grown considerably such that the CubeSat Design Specification [3] now defines 5 standardized form factors. The 1U CubeSat is the smallest form factor and represents the original design of a 10 cm cube with a mass of 1.33 kg. The other form factors use the 1U CubeSat as a base unit to define their respective dimension and mass requirements. The popularity of the CubeSat specification has allowed the 1U CubeSat to proliferate as a base unit of larger satellite designs as seen in Fig 1.

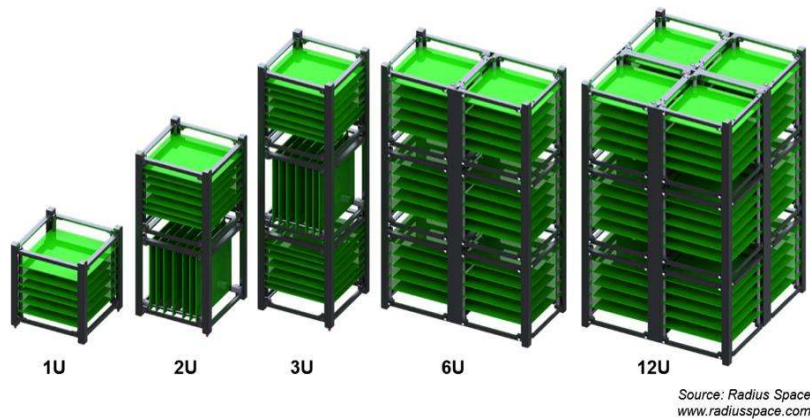


Figure 1: 1U, 2U, 3U, and 6U are standardized formats while 12U is a custom format [4]

A 3U CubeSat was chosen to house the payload within the rocket. This provides a structure to protect the payload during flight and an interface between the payload and the airframe of the rocket [2]. By following a common standard, additional learning opportunities are available to students. Furthermore, the SDL Payload Challenge incentivizes CubeSat development through bonus points [2].

1.2 Vibration Testing and Analysis

Vibration is mechanical oscillatory movement described by its amplitude and frequency [5]. The simplest form of vibration is a periodic sinusoidal motion or applied force [5]. In real environments, vibration is typically a combination of multiple sinusoidal functions and may be complex (not periodic) and random (unpredictable) [5].

Mechanical bodies and systems have modes at which they tend to vibrate at. Modes are characterized by simple harmonic motion of every particle at the same frequency [5]. Modes are described by their modal frequency and their mode shape. Real bodies and systems have an infinite number of modes [6].

Resonance is a physical phenomenon where the transmissibility of an applied vibration is increases as the frequency and shape of the applied vibration approaches that of a mode. Transmissibility describes a mechanical system's damping or amplification of an applied vibration. Fig. (2) graphs the transmissibility of a generalized mechanical systems with different critical damping factors, ζ , as a function of the ratio of the applied frequency (forcing frequency) to the modal frequency (natural frequency). In a minimally damped system where $\zeta = 0.01$, at resonance, vibration is amplified by over 4000%.

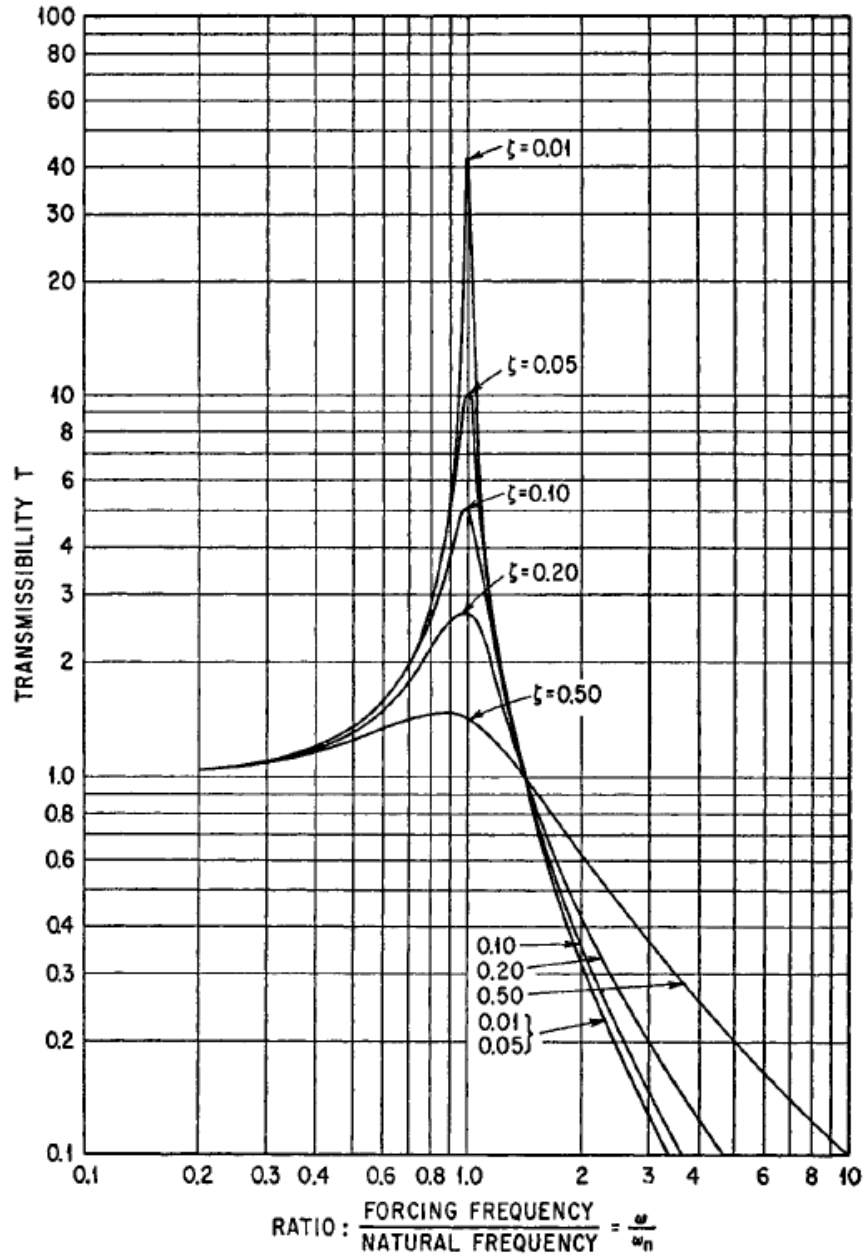


Figure 2: Vibration transmissibility with respect to ratio of forcing frequency to natural frequency, for varying damping ζ [5]

In aerospace, vibration is an important parameter to account for in design. The vibration experienced by a payload in a launch vehicle (LV) comes from mechanical sources and acoustic sources [7]. Rocket engines are particularly loud and are thus a source of violent acoustic vibration. Aside from backing out threaded fasteners and disconnecting electrical components, vibration is associated with fatigue [8]. If vibration is left unaccounted for, resonant behaviour results in material failure by accelerating the fatigue loading in ductile materials or amplifying the maximum stress in brittle materials.

To ensure that the concerns with vibration has been addressed in aerospace products, government bodies and aerospace agencies have specified design processes and testing standards for engineers to follow [7], [9]. When possible, vibration testing is conducted with shaker tables using measured vibration data from the LV that the product will fly in [7].

Vibration data is collected using accelerometers. From this data, the acceleration spectral density (ASD) is calculated. An ASD is used to characterize vibration environments as random vibration is a statistical event. Various software packages exist for calculating ASDs. In this project, the enDAQ VibrationData Toolbox [10] is used as it is available at no cost and is new user friendly.

With the base ASD of the LV environment calculated, the maximum predicted environment (MPE) can be derived. The MPE describes the highest ASD expected with 95% probability with 50% statistical confidence [7], [11]. To determine the MPE, an ASD envelope is created by encompassing the measured ASD data; it is permissible to neglect narrow bandwidth peaks as they contribute very little to the total vibration energy [11]. The MPE is created by raising the ASD envelope by 5 decibels (dB) as shown in Fig. 3. Changes in ASDs are measured in dB and can be calculated with the formula

$$N_{dB} = 10 \left[\log_{10} \left(\frac{W_2}{W_1} \right) \right]$$

where N_{dB} is the ASD change in dB, W_2 is the final ASD, and W_1 is the initial ASD [11].

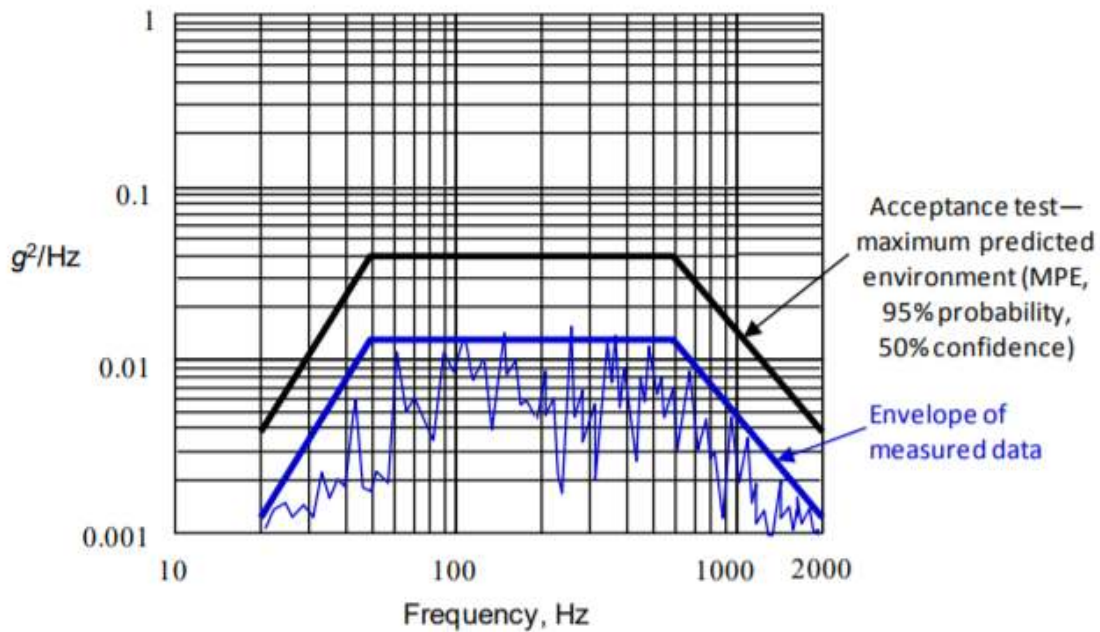


Figure 3: ASD envelope and MPE from measured ASD data [11]

From the MPE, the testing specifications for qualification, protoqualification (protoflight), and acceptance can be set. Qualification testing is for testing of dedicated hardware such that a *design* as tested. This is the most strenuous test standard and if passed, further flight hardware of the same design need only pass acceptance testing, which is the least strenuous test standard [5]. In cases where it is desirable to perform qualification testing on flight hardware, the protoflight test standard is used. The payload will be held to this test standard as only one will be manufactured and flown. The NASA protoflight standard specifies a test level of 3 dB above the MPE, applied for 60 seconds to each axis, individually [7], [11].

Typical design criteria for a CubeSat's first modal frequency require it to be greater than 100 Hz [12], [13]. This is intended to decouple LV modes from the CubeSat modes to prevent resonant behaviour; the modal frequencies of rockets are typically within the 5 Hz – 100 Hz bandwidth, often favouring the lower end of that range [14]. Modal analysis is used to identify potential dynamic coupling. Modal analysis describes the dynamic response of a system by determining its modes.

Real bodies have an infinite number of modes but typically, only a small subset of those modes is relevant to the dynamic response of a system. To determine which modes are relevant and when the dynamic response of a system is sufficiently characterized, the effective mass participation factor (EMPF) and cumulative effective mass participation factor (CEMPF) are used, respectively. EMPF describes the amount of energy involved in a mode and CEMPF is the cumulative total of EMPF [15]. Thus, modes with a high EMPF contribute significantly to the dynamic response of a system. EMPF and CEMPF are measured on a percent scale and are axis specific. To sufficiently characterize the dynamic response in an axis, the CEMPF must be > 80% [15] while practice in aerospace trends towards > 90% [12]. Additionally, aerospace vibration testing is typically bounded between 20 Hz and 2000 Hz [11], [13].

1.3 Ansys and FEA

Ansys is a popular finite element analysis (FEA) software used to simulate the physical behaviour of engineering models [16]. FEA is based on the finite element method (FEM) where physical bodies are discretized into a finite number of elements. The behaviour of these elements is numerically described using matrices.

Ansys was chosen for this project due to its user-friendly features, wealth of online documentation, and large user base. This means that a new user with little FEA experience may quickly learn how to use the software and leverage the tools that Ansys offers. Additionally, Waterloo Rocketry has access to Ansys licences which allow access to the full suite of Ansys analysis software. FEA is a common design tool in the aerospace industry that is used to evaluate designs before production. This aligns with the overall objectives of Waterloo Rocketry of providing students with relevant experience and skills.

The process of running an Ansys simulation is as follows: 1) create model geometry, 2) preprocess model, 3) solve model, and 4) postprocess model [17].

Step 1 is completed in any commercial computer aided design (CAD) software; Waterloo Rocketry uses SolidWorks by Dassault Systemes.

Step 2 is where the CAD model is prepared for FEA. In this step, the model is simplified, meshed, and constrained. FEA is a computationally expensive and intensive process. Simplifications and approximations increase success and decrease runtime. The simplified model is then meshed. This is when the model is discretized into finite elements. Constraint equations are then applied through contacts that relate the meshes of different bodies with each other and the ground. Applied loads and environmental conditions are set at the end of this stage.

Step 3 is where the FEA software takes over and solves the simulation. Simulation run times can vary from half a minute to days depending on the complexity of the model.

Step 4 is where the FEA results are analyzed and made intelligible. Different plots may be generated showing deformation, stress, and other details.

In this project, Ansys Static Structural, Modal Analysis, and Random Vibration Analysis are used to evaluate the payload. Static Structural is used to simulate quasi-static loading. Modal Analysis is used to determine the modes of a structure. Random Vibration Analysis uses the results from a Modal Analysis and a user defined ASD function to simulate the dynamic response of the model to the given ASD [16].

1.4 Current Payload Design

The payload is a 3U CubeSat housing a systems module, photoplates module, and scintillator module [2]. The systems module controls and powers the payload. The photoplates module contains a stack of coated glass plates which show the paths of ionizing radiation after being developed, like photographic film. This module is still in the early stages of development and significant changes may occur. The scintillator module is the primary experiment of the payload. Its objective is to measure the radiation shielding of boron nitride nanotube-thermoplastic polyurethane and metal oxide-acrylic composites. The systems module is in the bottom, the photoplates module is in the center, and the scintillator module is in the top. The current structure of the CubeSat is shown in Fig. 4.

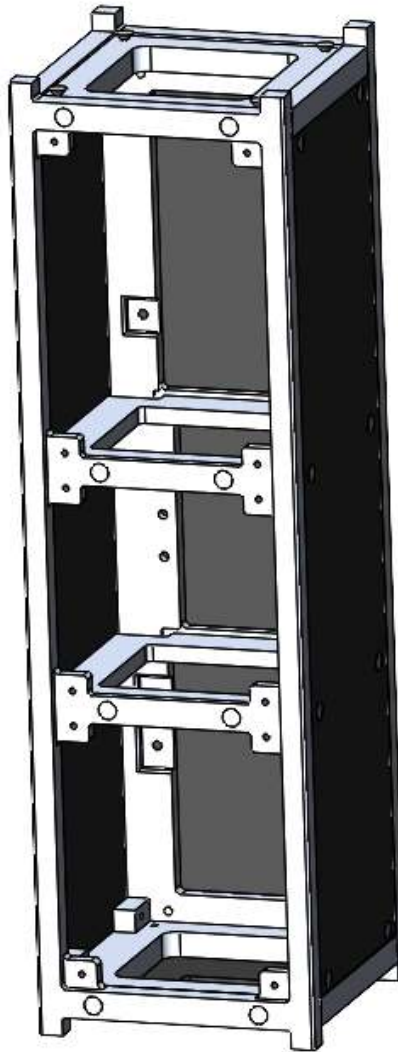


Figure 4: Current CubeSat design, without internal modules and front cover panels

The CubeSat structure follows a conventional design. It consists of two major side faces, held together by braces, and sealed off with thin cover panels. This design features many open areas and cut-outs. Waterloo Rocketry has not flown or mechanically tested a CubeSat design of this nature and thus the flightworthiness of the current design is unknown.

Though the design is subject to change, the general layout of the structure is not expected to change. Further details regarding the design of the CubeSat structure may be found in Tam's work report [2].

1.5 Objective

The objective of this project is to ensure flightworthiness of the payload. This includes the CubeSat and everything housed within it for flight aboard the Kraken of the Sky (KOTS), a hybrid engine rocket targeting an altitude of 30 000 feet. FEA analysis is applied to verify flightworthiness of the payload in static and dynamic loads intended to replicate the environment within KOTS. Components and systems found to be non-flightworthy are identified and corrected according to FEA results. Time, resource, and risk permitting, the payload will be subject to random vibration testing.

The secondary objective of this project is to develop procedures for ensuring payload flightworthiness. The intent is for analysis projects like this one to become standard practice in subsequent payload design cycles.

This report is created to document the state of the project and analysis of the 2021 payload. The next steps in this project are recorded at the end of the report.

2.0 Problem Definition

Waterloo Rocketry has the need for a flightworthy payload, capable of safely housing the 2021 payload experiment, to ride aboard a full altitude flight of KOTS. The payload must pass inertial and dynamic load testing conditions that are set to approximate the environmental conditions of a full altitude flight inside KOTS' payload bay. Design changes taken to improve flightworthiness must minimize added manufacturing and assembly complexity.

2.1 Applicable Standards

The same applicable standards listed in [2] will apply here, in the same order of precedence. Several additional standards have been added to the hierarchy to reflect the potential of competing in the Launch Canada Challenge and to complement the scope of this project.

The Design, Test, & Evaluation Guide by Launch Canada Challenge [18] will hold the same precedence as the SA Cup Rules and Regulations [19]; both standards are similar as the Design, Test, & Evaluation guide is heavily influenced by the SA Cup Rules and Regulations.

NASA technical standard NASA-STD-7001B (Payload Vibroacoustic Test Criteria) and the Tohnichi Torque Handbook are added to reflect the technical scope of the project.

All the applicable standards are listed in Table 1 in descending order of precedence.

Table 1: Applicable standards, in descending precedence [2]

STANDARD	JURISDICTION
SA Cup Rules and Regulations [19]	Experimental Sounding Rocket Association
Design, Test, & Evaluation Guide [18]	Launch Canada Challenge
CubeSat Design Specification [3]	Waterloo Rocketry (optional for competition)
Waterloo Rocketry CAD Standard [20]	Waterloo Rocketry
NASA-STD-7001B [7]	Waterloo Rocketry (optional for competition)
Tohnichi Torque Handbook [21]	Waterloo Rocketry (optional for competition)

2.2 Timeline

The overarching objective, beyond that of this project, is to fly an experimental payload aboard KOTS at either the Launch Canada Challenge or SA Cup. Both events are anticipated to take place in the Spring 2021 term. Thus, manufacturing of the payload must be complete by early May 2021. Simulation for flightworthiness verification will occur on an *ad hoc* basis as components and designs are finalized.

The CubeSat structure is expected to, at most, see dimensional changes as the design of the internal modules mature. Thus, it will be of first focus in flightworthiness verification. The CubeSat structure shall be verified by early February.

The internal modules shall be flightworthiness tested at the module level when their designs are sufficiently mature. Ideally, testing at the module level will be completed prior to April such that sufficient time is left to freeze the design to allow for drafting and fabrication.

2.3 Constraints and Criteria

Constraints and criteria will include and expand upon those set in [2]. Details regarding the payload (CubeSat and internal modules) are unchanged from [2]. Design modifications made through the course of this project will follow those set constraints and criteria. In addition, design changes to the payload shall increase flightworthiness. A flightworthy payload is described by the following attributes: a first modal frequency above 100 Hz, where higher is better; no modes likely to significantly damage the payload; and a factor of safety of at least 1.5 in every test case [2].

Additional considerations specific to this project pertain to the Ansys FEA aspect. Simulations are to be kept small to reduce runtime, while still providing valid results. It is easy to generate an overly complex model that does not provide more analytical value than a simpler alternative. A complex model may take orders of magnitude longer to run than a simple model. Thus, there is a concentrated effort to simplify and approximate models where appropriate.

3.0 Technical Progress

The project has been broken down into a series of phases where the detail of the model and complexity of analysis increase with each iteration. Analysis starts with a simple, arbitrary load applied to a single body, and builds to a full assembly with inertial loading and modal analysis.

3.1 Stepped FEA Approach

The end objective is to model the entire payload within Ansys. A complete model would feature many details and numerous parts that would all need to be correctly preprocessed for the solver to succeed. Juggling all the details in one sweep would be daunting for an inexperienced Ansys user and likely result in a frustrating and time-consuming experience. Therefore, a three-phase approach is taken to gradually build up the model to its desired level of complexity. By gradually introducing complexity, problems will be easier to diagnose and resolve.

Phase 1 is to familiarize with the Ansys FEA workflow. To do this, a single part is analyzed. The emphasis is on developing good meshing practice and familiarity with the graphical user interface.

Phase 2 is to build up complexity and analyze the static and modal behaviour of a simplified CubeSat structure. The emphasis is on reinforcing good preprocessing practice and investigating some of the different parameters and settings that can be adjusted in the analysis system.

Phase 3 is to build a near full detail model and conduct flightworthiness testing with the static and dynamic loads expected inside KOTS.

3.2 Progress to Date

Phases 1 and 2 have been completed. Phase 3 is partially complete.

3.2.1 Phase 1

In phase 1, the front face of the CubeSat structure was chosen for analysis. A simplified model was compared against a full detail model. Body fitted cartesian, hex dominant, and tetrahedral meshing methods were used. Static structural and modal analysis was conducted on the models.

To simplify the front face, all radii and countersinks were removed. Threaded holes were retained. The full detail model was left unchanged from the CAD model.

The simplified model was meshed with the body fitted cartesian and hex dominant methods. The full detail model was meshed with the tetrahedral method; the countersinks and radii made a hex

dominant mesh with sufficient granularity difficult, while a body fitted cartesian mesh would not accurately retain the geometry. The resultant meshes are shown in Fig. 5, Fig. 6, and Fig. 7.

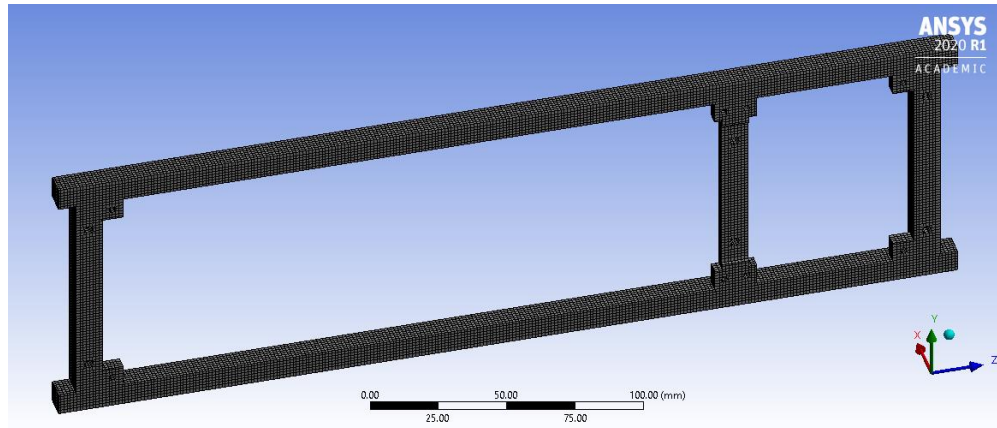


Figure 5: Body fitted cartesian mesh of simplified CubeSat front face

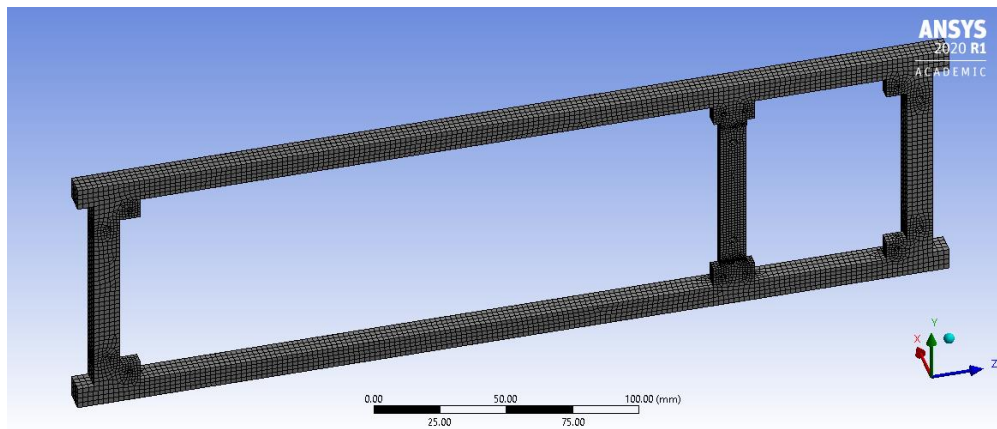


Figure 6: Hex dominant mesh of simplified CubeSat front face

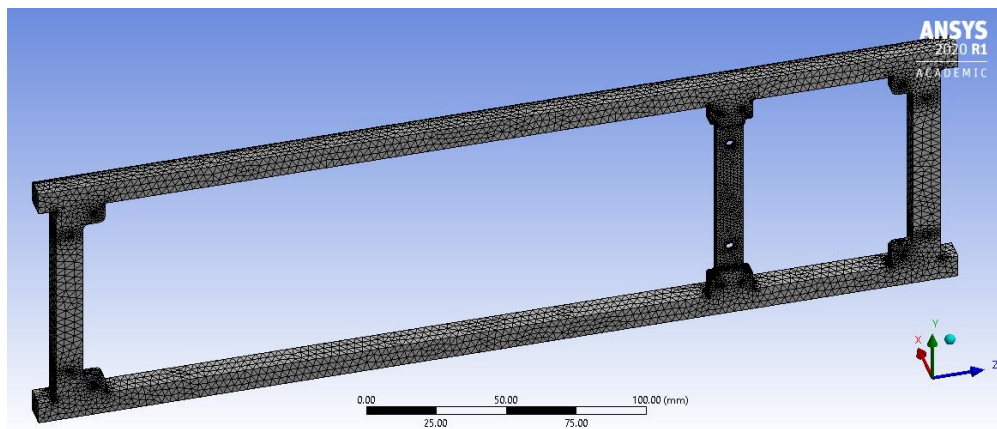


Figure 7: Tetrahedral mesh of full detail CubeSat front face

Typical meshing rule-of-thumb dictates a minimum of three elements between faces to accurately capture behaviour. To achieve this, a variety of global and local mesh controls were applied.

Adaptive sizing was turned off and proximity and curvature controls were used; proximity and curvature controls are referred to as advanced sizing functions (ASF). Proximity specifies a minimum number of elements between faces and edges while curvature specified the maximum angle that a single element can span. Fig.8 shows how proximity and curvature controls effect a surface mesh. Face sizing was the only local mesh control used in phase 1.

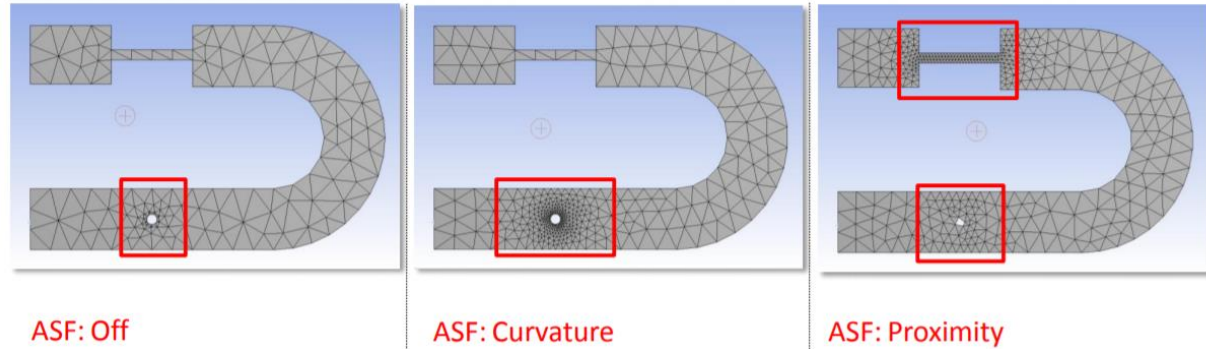


Figure 8: Effects of curvature and proximity advanced sizing functions (ASF) [17]

Mesh element quality statistics are used to measure the quality of a mesh. Poor mesh quality can result in inaccurate analysis results. Different metrics exist for measuring element quality such as Jacobian ratio, skewness, and aspect ration, among others [22]. Ansys provides an *element quality* metric which factors in these different metrics. The higher the *element quality*, the more “ideal” an element is. A higher *element quality* is desirable.

In all three models, the Ansys data for aluminum 6061 T6 was applied. Similarly, the bottom faces of the bottom feet of the front face were fixed to the ground. Static structural analysis in each case consisted of a 500 N compressive load applied to the top faces of the top feet. Modal analysis of each model was conducted. The results of each analysis are tabulated in Table 2.

Table 2: Phase 1 front face FEA mesh and results comparison

MESH METHOD	BODY FITTED	HEX	
	CARTESIAN	DOMINANT	TETRAHEDRAL
MODEL DETAIL	Simplified	Simplified	Full detail
NUMBER OF MESH NODES	184 791	84 578	181 009
MESH ELEMENT QUALITY AVERAGE	0.989	0.779	0.807
MESH ELEMENT QUALITY STANDARD DEVIATION	0.032	0.242	0.108
STATIC STRUCTURAL MAX STRESS (MPA)	1225	1052	1537
STATIC STRUCTURAL COMPUTATION TIME (S)	49.3	23.4	49.0
FIRST MODAL FREQUENCY (HZ)	374.56	376.15	376.11

The first observation is that a hex dominant mesh can be significantly smaller which allows the analysis system to solve in less than half the time it takes for the larger body fitted cartesian and tetrahedral meshes. Another observation is that the modal results between all three models vary insignificantly.

However, of concern is the wide range of max stress results. It is to be noted that between the simplified and full detail models, the location of max stress varies. In the full detail model, the location of max stress is in the fillets of the bottom feet while in the simplified models, the location of max stress is in a corner near the top of the front face. With a change in location and an up to 46% increase in magnitude, the simplified and full detail models behave significantly differently.

These results indicate the benefits and detriments of simplification. With a simplified model, a more efficient mesh can be generated using a swept or hex dominant method. This can drastically decrease solution time. However, through simplification, important details may be lost and the FEA model may no longer be representative of the CAD model. This shows the critical role that simplification plays in FEA.

3.2.2 Phase 2

In phase 2, a simplified CubeSat structure was modeled for analysis. This model consisted of the five major structural bodies and their respective fasteners, as seen in Fig. 9. The cover panels and internal models were omitted in this phase of analysis. Hex dominant and tetrahedral meshed models were generated and compared. Static structural and modal analysis was conducted on both models.

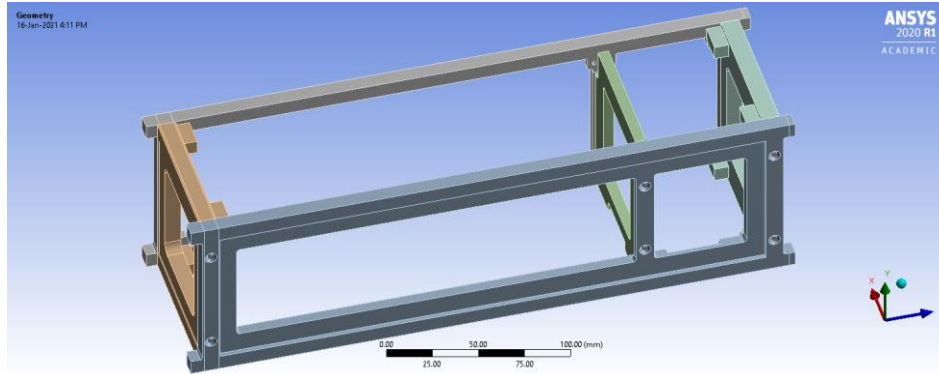


Figure 9: Phase 2 FEA model, with simplifications applied

Simplifications taken included removing chamfers, removing rail radii, and applying virtual topology to small faces not of interest. To further simplify the model, body-body beam connections were used to represent the 4-40 screws in the structure. The beam bodies were set to a diameter of 2.8448 mm (0.112 in.). The material properties of austenitic stainless steel were applied to the beam bodies. The structural bodies were set as aluminum 6061 T6.

For clarity of discussion, “up” relative to the CubeSat is in the -Z direction, and consequently “down” is in the +Z direction, as defined by the coordinate system in Fig. 9.

Hex dominant and tetrahedral meshes were generated for this model. The hex dominant mesh method required a significant amount of user time to detail parameters for a suitable mesh. Proximity and curvature ASF controls were applied at the global level. A variety of body, face, and edge sizing controls were applied in addition to face meshing at the local level.

Significant knowledge and experience are required to generate a hex dominant mesh; new users will find hex dominant meshing extremely frustrating due to the lack of specificity in error messages. The hex dominant mesh is shown in Fig. 10. The tetrahedral mesh was significantly easier to generate; tetrahedral meshing is less prone to failure. The tetrahedral mesh is shown in Fig. 11.

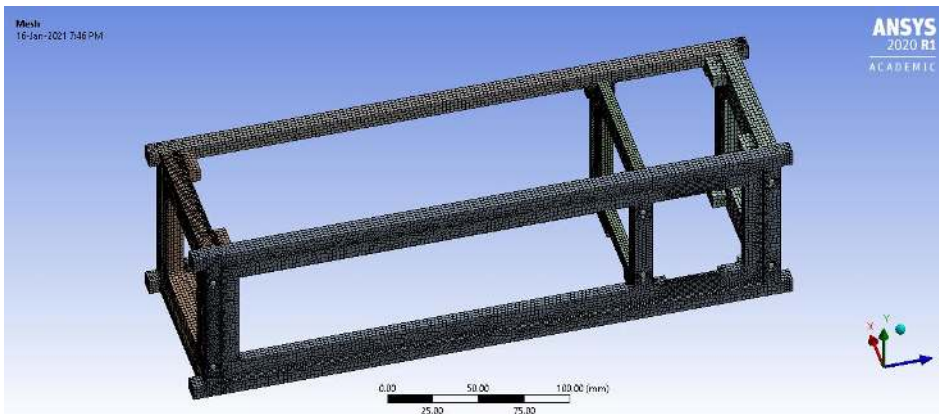


Figure 10: Phase 2 hex dominant mesh

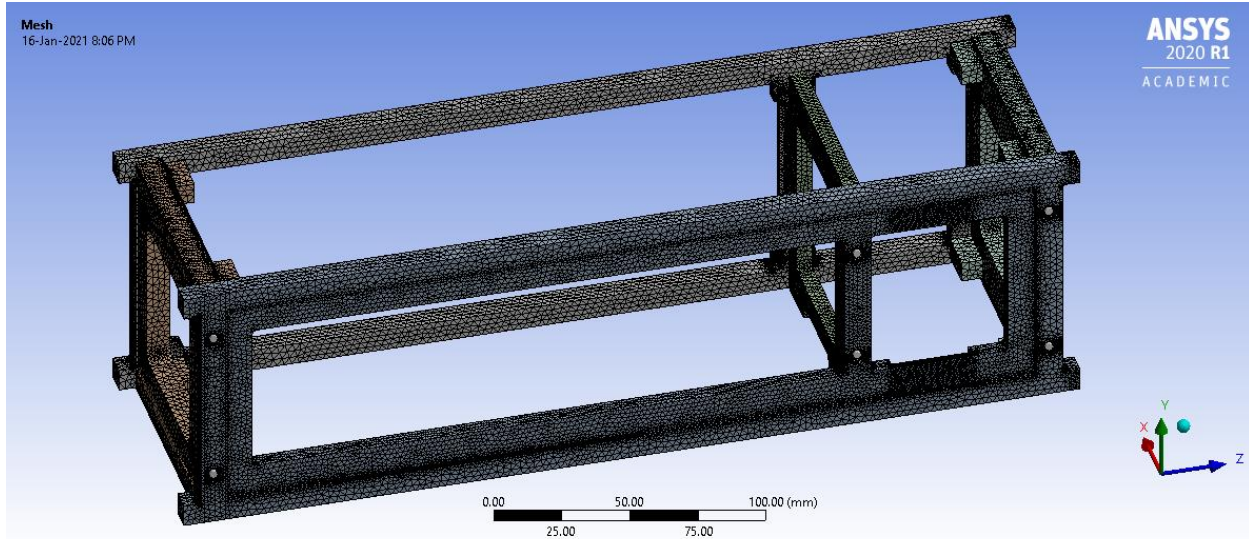


Figure 11: Phase 2 tetrahedral mesh

With multi-body meshes, the interactions between the bodies must be defined. Linear contact types such as the fixed contact type are easier to solve but may not accurately represent the system. Thus, the nonlinear frictional contact type was chosen to represent the interactions between the bodies. It is assumed the aluminum-on-aluminum contact surfaces are clean and dry. For such a contact type, the coefficient of sliding friction is 1.4 [23], [24]. The coefficient of static friction varies between 1.05 - 1.35 [23] so the mean of 1.2 is chosen.

To accurately determine the static and dynamic behaviour of the payload, the payload bay (PayBay) that the CubeSat is mounted in must be represented in the FEA model. Fig. 12 shows the current PayBay design. The PayBay consists of upper and lower braces that are fixed in place within KOTS' body tube with epoxy. The upper brace constrains motion in the transverse direction of the rocket. The lower brace constrains motion in the transverse direction, and vertically braces the bottom of the CubeSat. A 4-40 screw is threaded into the upper portion of the CubeSat to fix the CubeSat axially. Serendipitously, all major axial loading on the CubeSat is predicted to occur in the same direction such that the lower brace will support all the major load cases. The materials of these braces are not finalized, but may be either fiberglass, plywood, or aluminum.

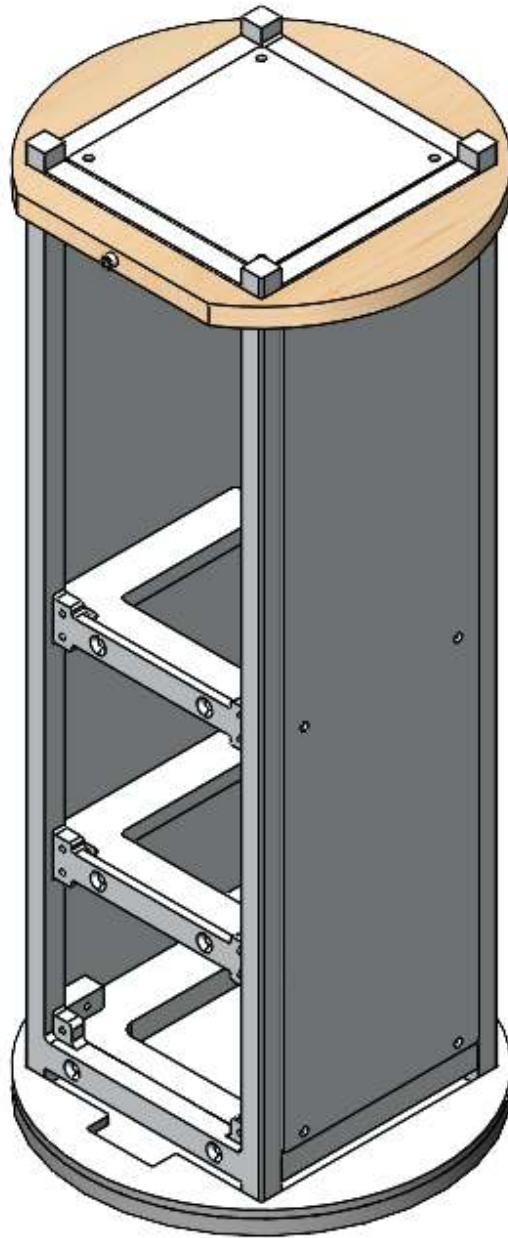


Figure 12: Payload bay (PayBay) with CubeSat installed

To model PayBay within Ansys, additional faces were specified for compression only supports, as detailed in Fig. 13. An 8 mm wide annular ring around the CubeSat in the X-Y plane was specified for the upper brace. The lower brace was scoped to the bottom faces of the bottom feet and a 3.175 mm wide ring applied to the outside faces of the bottom feet. The threaded 4-40 hole was set as a fixed support. These braces are assumed to be rigid for this project. In future work, it is worthwhile to consider modelling the deflection of these bodies.

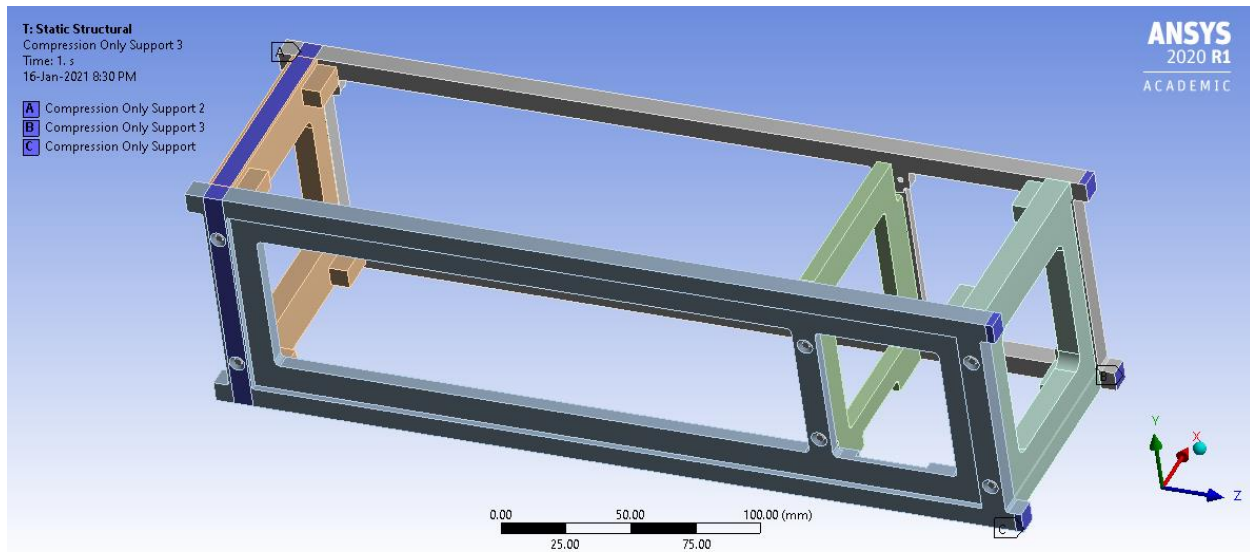


Figure 13: CubeSat compression only support faces

Static structural and modal analysis was run on both the hex dominant and tetrahedral meshes. The static structural load was defined as a 12G acceleration in the -Z direction where G is Earth's gravitational acceleration defined as 9.81 m/s^2 .

The analysis results of both the hex dominant and tetrahedral meshes were comparable. The details of such are in Table 3. It is noted that the tetrahedral mesh was significantly easier to generate and work with than the finicky hex dominant mesh that was prone to fail during mesh generation. In addition to greater difficulty in meshing, the hex dominant mesh was also substantially inferior in average element quality in comparison to the tetrahedral mesh. However, by being a more efficient mesh with 34% less nodes, the hex dominant mesh solved 32% faster than the tetrahedral mesh.

Table 3: Phase 2 simplified CubeSat FEA mesh and results comparison

MESH METHOD	HEX DOMINANT	TETRAHEDRAL
NUMBER OF MESH NODES	493 763	745 640
AVERAGE MESH ELEMENT QUALITY	0.644	0.814
MESH ELEMENT QUALITY STANDARD DEVIATION	0.252	0.101
STATIC STRUCTURAL MAX STRESS (MPA)	1.89	1.92
STATIC STRUCTURAL COMPUTATION TIME (MIN)	140	207
FIRST MODAL FREQUENCY (HZ)	425.21	425.63

The static and dynamic results varied insignificantly. Maxima of stress were in the PayBay screw hole and radii of the bottom feet in both models. The first modal frequency of both models was also within 1 Hz of each other.

From these results, it was concluded that future meshes will be generated with the tetrahedral method. The simulation runtime savings were not worth the substantial difficulties encountered when using the hex dominant mesh method.

3.2.3 Phase 3

In phase 3, a complete payload model is analyzed. This model expands upon the work from phase 2 to include the cover panels and internal modules. The model is tetrahedrally meshed and initial flightworthiness analysis is conducted.

Simplifications include removing chamfers and removing rail radii. Like in phase 2, body-body beam connections are used to model the 4-40 screws in the structure. Frictional contacts are used with the same coefficients as specified in phase 2 to model the contact surfaces between all parts. It is to be noted that gaps should be avoided with frictional contacts when possible; gaps in frictional contacts prevent the solver from converging. It is assumed that the faces along the thickness of the aluminum cover panels will not contact the main structure. This is because of the locating properties intrinsic to countersunk screws. Thus, no contacts are defined for the faces along the thickness of the cover panels.

The internal module designs are yet to be finalized. The current designs are volatile and CAD models are not available. Thus, the internal modules are approximated. Each of the three internal modules (systems, photoplates, and scintillator modules) are represented by a point mass located in the geometric center of their respective module bays. The point masses are attached to the CubeSat through the respective threaded holes of each module.

The current CubeSat structure is approximately 1500 g, the systems module 1099 g, the photoplates module 483 g, and the scintillator module 500 g. This sums to 3582 g. The payload must have a mass of 4000 g to meet competition requirements. To meet this mass requirement, deadweight is added to the modules. Of note, CubeSat specifications call for the center of mass of the CubeSat to be within 20 mm of its geometric center in the X and Y axis and within 70 mm in the Z axis [3].

In this analysis project, the roughly 400 g difference between current approximated mass and target mass is distributed among the three modules as an additional mass budget, intended to account for mass increases due to design changes. The allocation of mass budget favours the photoplates module as its design is the least mature of the three modules. As such, the systems module is simulated with a mass of 1200 g, the photoplates module a mass of 700 g, and the scintillator module a mass of 600 g.

In addition to the inclusion of internal modules, screw preloads are added to the analysis as they substantially change the dynamic behaviour of the system [8]. 4-40 screws of either stainless steel or alloy steel construction are threaded into the aluminum CubeSat structure. It is expected that the aluminum threads fail before the screw shank when the joint is loaded in tension. Thus, in reference to the Tohnichi bolt tightening handbook [21], torque preload values are taken from the 0.5T series. The handbook specifies preloads for metric size screws. M3 is the metric closest equivalent to #4 UNC screws. The 0.5T series M3 axial preload is specified at 528 N [21]. To account for the slightly smaller nominal diameter of the #4 screw and to round the screw preload to the nearest hundred Newtons of force, the screw preloads are 500 N.

Static loading due to inertial acceleration and dynamic loading due to LV vibrations are used to determine preliminary flightworthiness of the payload in this phase of analysis. Flight data for the previous rocket Shark of the Sky (SOTS) and simulation data for KOTS are used to determine the test specifications. Static and dynamic loading are highest during launch of a rocket [11]. Loading due to parachute deployment is calculated to be no more than 2G. Thus, test conditions will be based on data during launch.

Of greatest concern is the behaviour of the payload under axial inertial loading and tri-axial random vibration loading. Axial inertial loading includes the greatest acceleration, in the -Z direction, and the prominent shear failure mode of the PayBay screw, in the +Z direction. Tri-axial random vibration loading will follow protoflight standards defined in NASA-STD-7001B. These load cases take priority.

Secondary priority load cases are the transverse inertial loads in the Y, +X, and -X directions. The model is symmetric about the X-Z plane, thus analysis in the +Y and -Y directions can be simplified to the Y direction.

Accelerometers aboard SOTS measured the axial and transverse acceleration over the course of its flight. Axial acceleration data from SOTS is plotted with respect to time in Fig. 14. The maximum recorded acceleration is a peak of 23.3G. This may be partially attributed to sensor noise or the mounting of the accelerometer. The majority of recorded acceleration is at or below 15G. Openrocket simulation data for KOTS reports a maximum acceleration of 7.1G. -Z axial inertial loading is set at 15G to envelope the majority of SOTS acceleration. Maximum negative acceleration recorded on SOTS is a narrow 5.1G peak. +Z axial inertial loading is set at 5G to envelope the majority of SOTS acceleration.

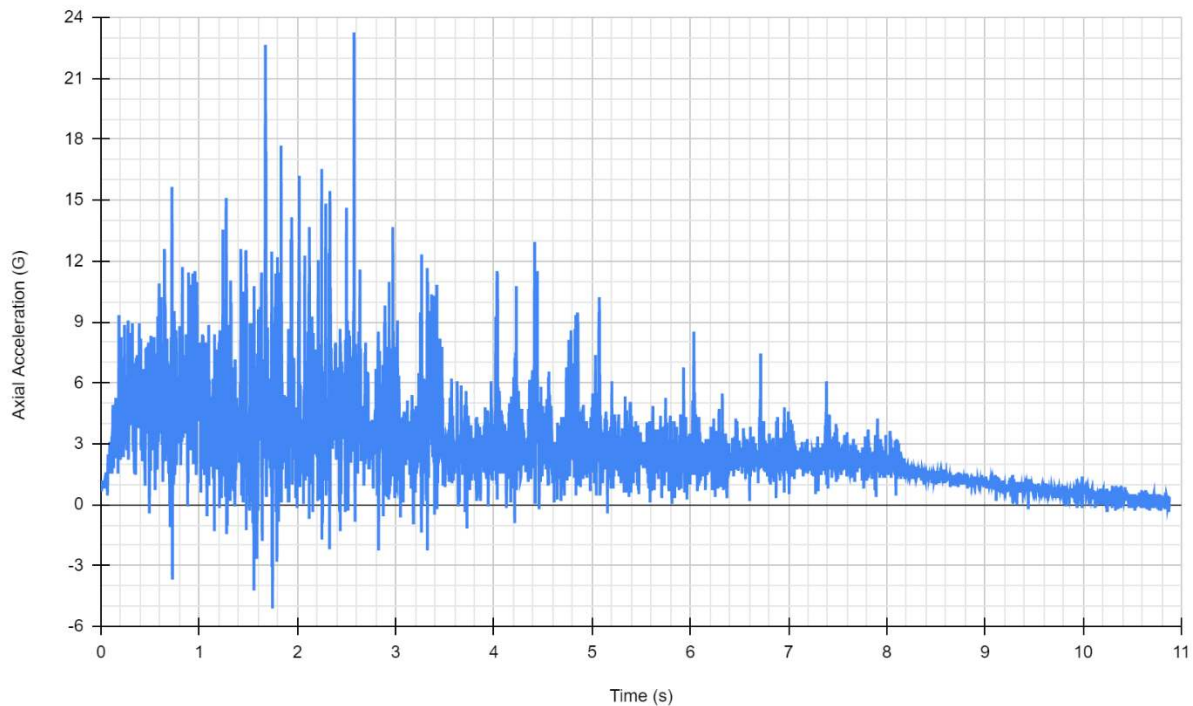


Figure 14: SOTS axial acceleration with time

Maximum transverse acceleration on SOTS is measured at 10.1G. There may be significant sensor noise. Maximum transverse acceleration from the openrocket simulation of KOTS is 1.3G. It is unknown how this value is calculated. The transverse (X and Y axis) inertial loads are set at 12G due to the uncertainty in data.

Random vibration analysis in Ansys is set with ASD plots. ASD protoflight testing plots are generated from SOTS acceleration data in accordance with [7]. The measurement rate of the accelerometer restricts the SOTS ASD plot to an upper limit of 200 Hz at best. Data is extrapolated aggressively and supplemented with the component minimum workmanship random vibration test levels [7] specified in Table 4.

Table 4: Component minimum workmanship random vibration test levels [7]

20 Hz	@	0.01 g ² /Hz
20 to 80 Hz	@	+3 dB/oct
80 to 500 Hz	@	0.04 g ² /Hz
500 to 2000 Hz	@	-3 dB/oct
2000 Hz	@	0.01 g ² /Hz
Overall Level	=	6.8 grms

Axial and transverse ASD plots for SOTS are generated with 3 Hz bin width with the VibrationData Toolbox [10] and shown in Fig. 15 and Fig. 16, respectively. 3 Hz bin width is used to increase readability of the plots [11]. ASD envelopes, MPEs, and protoflight ASDs are derived from these plots.

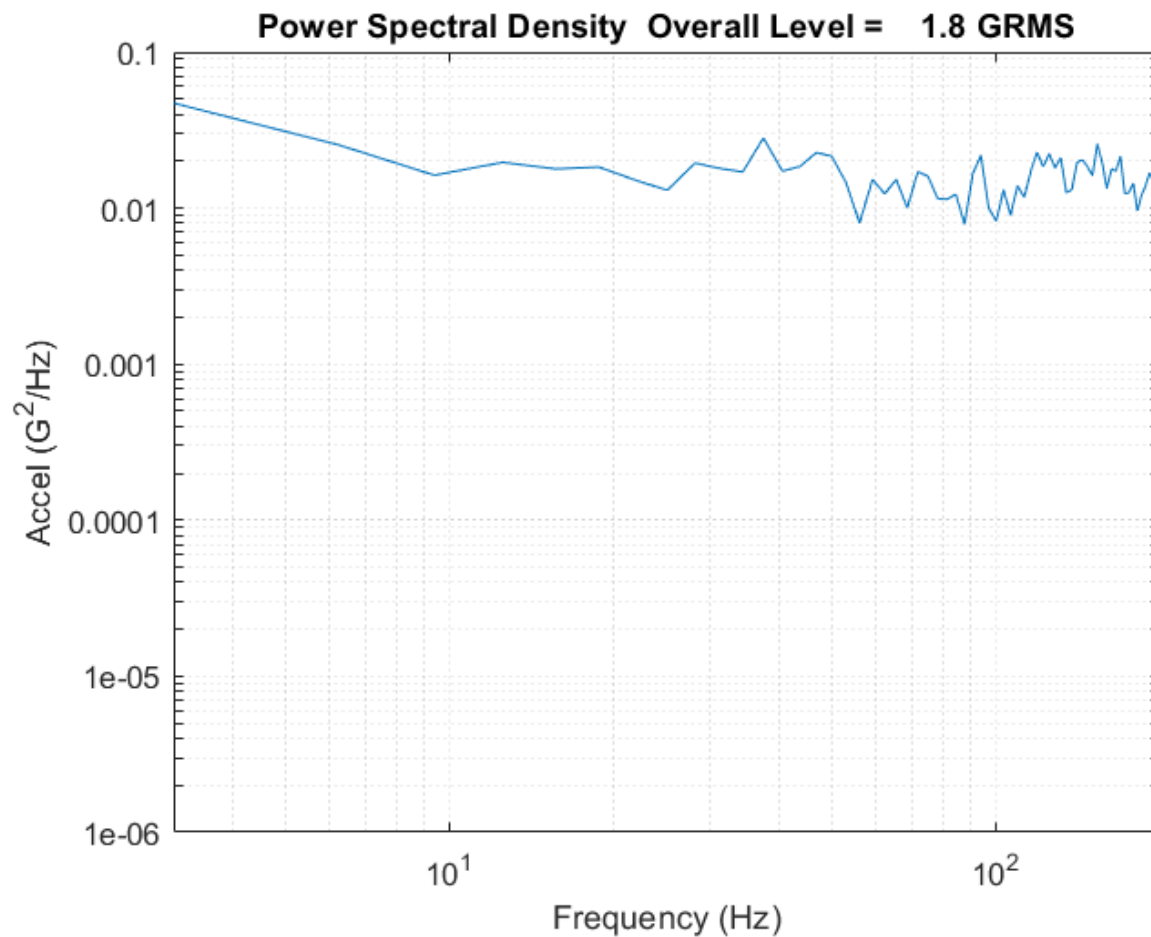


Figure 15: SOTS axial ASD with 3 Hz bin width

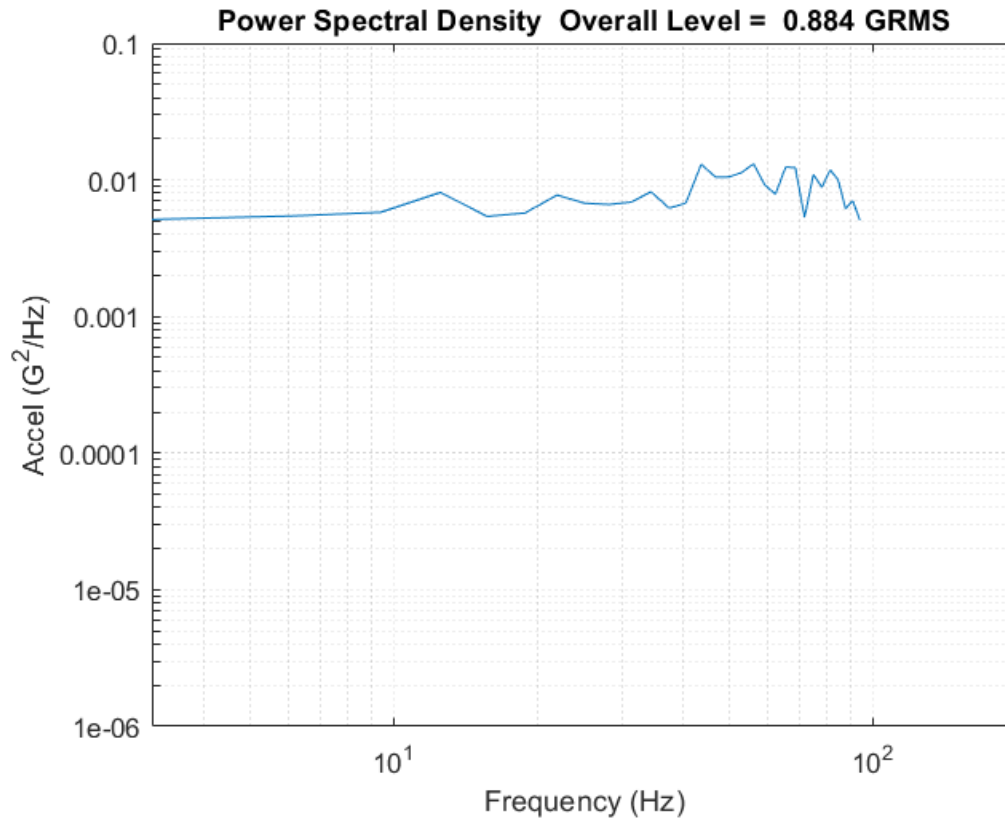


Figure 16: SOTS transverse ASD with 3 Hz bin width

The axial ASD envelope is drawn horizontally at $0.025 G^2/Hz$ from 20 Hz to 100 Hz. Increasing this value by 8 dB yields $0.16 G^2/Hz$, which is the protoflight ASD. This ASD value is extrapolated to 500 Hz, after which the plot decreases to $0.01 G^2/Hz$ at 200 Hz, in accordance with [7]. Fig. 17 shows the axial protoflight ASD.

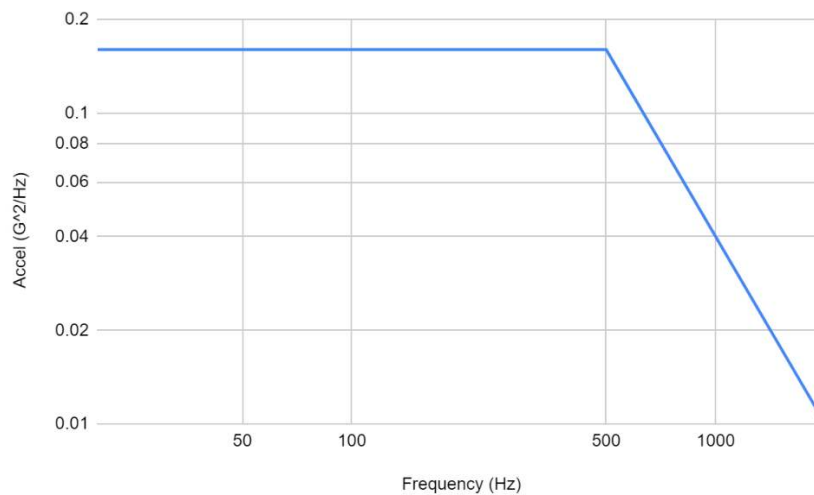


Figure 17: Axial protoflight ASD

The transverse ASD envelope is drawn horizontally at $0.02 \text{ G}^2/\text{Hz}$ from 20 Hz to 100 Hz. Increasing this value by 8 dB yields $0.13 \text{ G}^2/\text{Hz}$, which is the protoflight ASD. This ASD value is extrapolated to 500 Hz, after which the plot decreases to $0.01 \text{ G}^2/\text{Hz}$ at 200 Hz, in accordance with [7]. Fig. 18 shows the axial protoflight ASD.

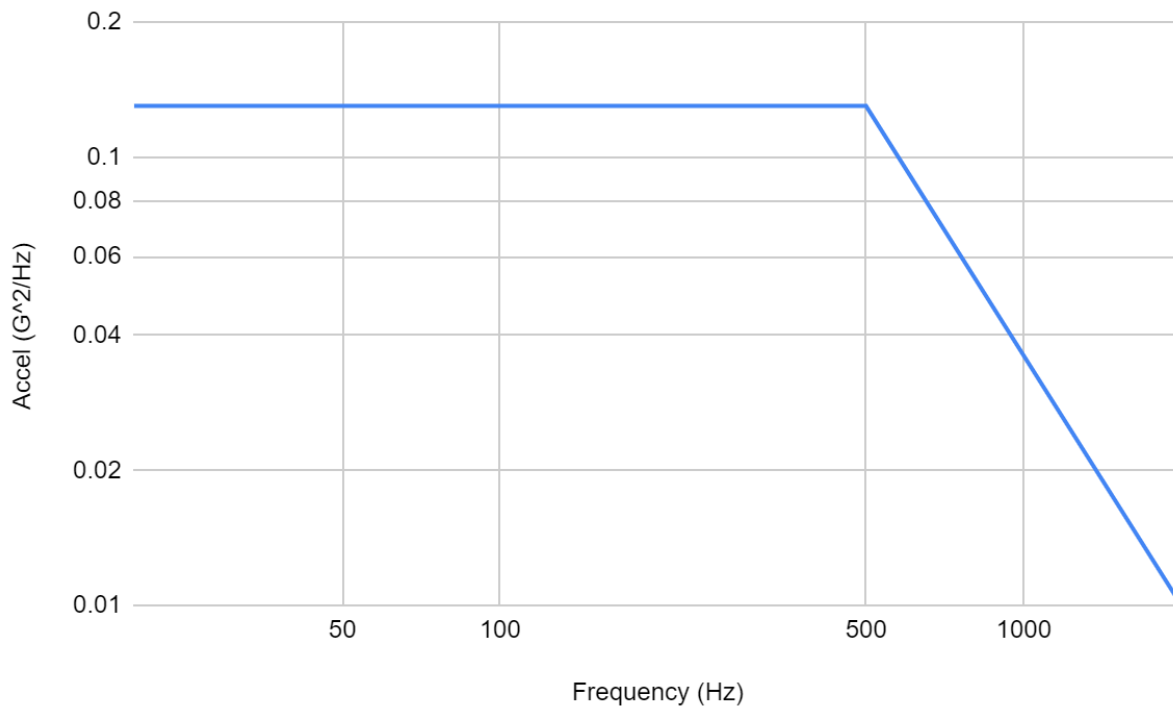


Figure 18: Transverse protoflight ASD

For random vibration testing, the axial protoflight ASD will be applied in the Z axis and the transverse protoflight ASD will be applied in the X and Y axis, simultaneously.

The model is meshed with the tetrahedral method. The mesh is shown in Fig. 19. To keep the number of elements low, the cover panels are coarsely meshed to be one element thick. The panels are not expected to bear load and modal behaviour is the primary interest. Thus, a coarser mesh is viable as granularity along the thickness of the panels is not needed. The mesh totals 553 337 nodes. The average element quality is 0.785 with a standard deviation of 0.118. Supports are carried over from phase 2. A fixed support is defined for the PayBay screw hole. Compression only supports are set for the PayBay contact surfaces.

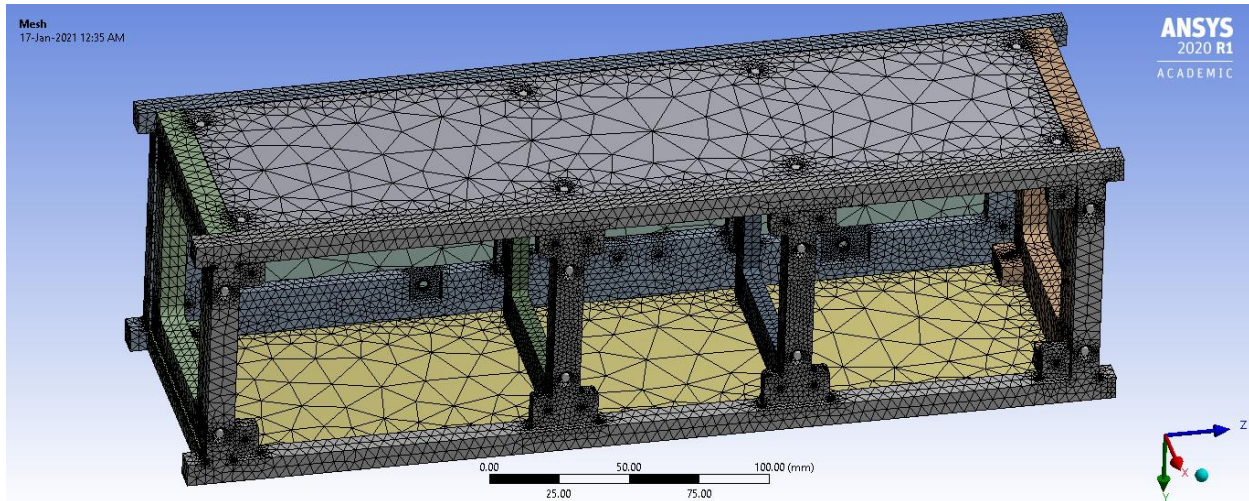


Figure 19: Phase 3 full model mesh

Inertial load analysis for -Z, +Z, and Y directions are conducted; Fig. 20, Fig. 21, and Fig. 22 show the max stress plots for each case, respectively. Inertial loads in the +X and -X directions still need to be analyzed.

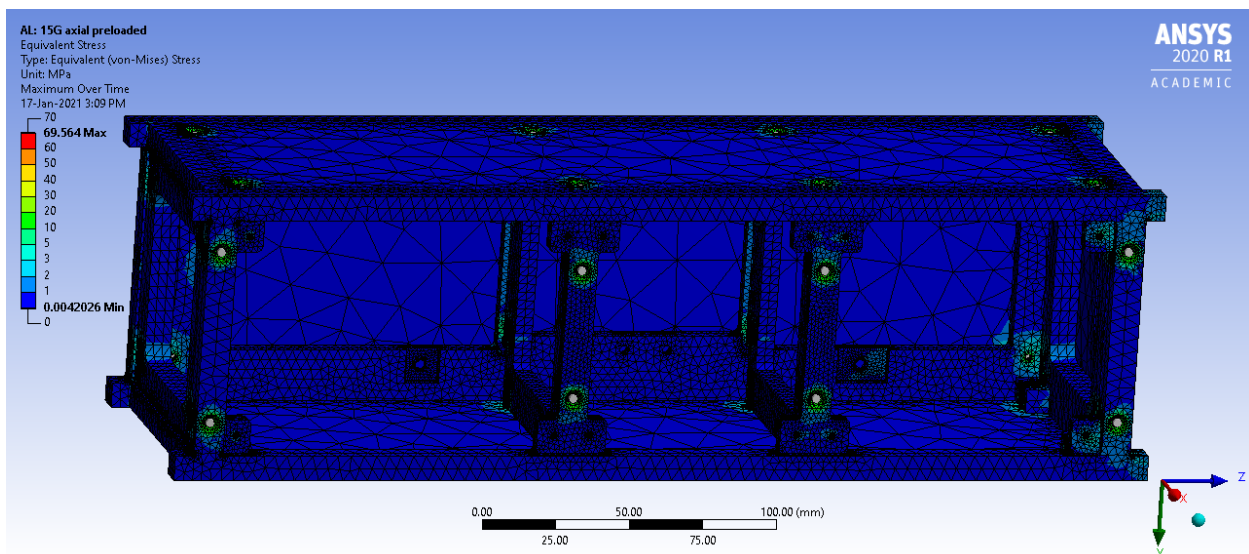


Figure 20: Phase 3 full model 15G -Z direction inertial load maximum stress

Figure 20: -Z direction inertial load max stress

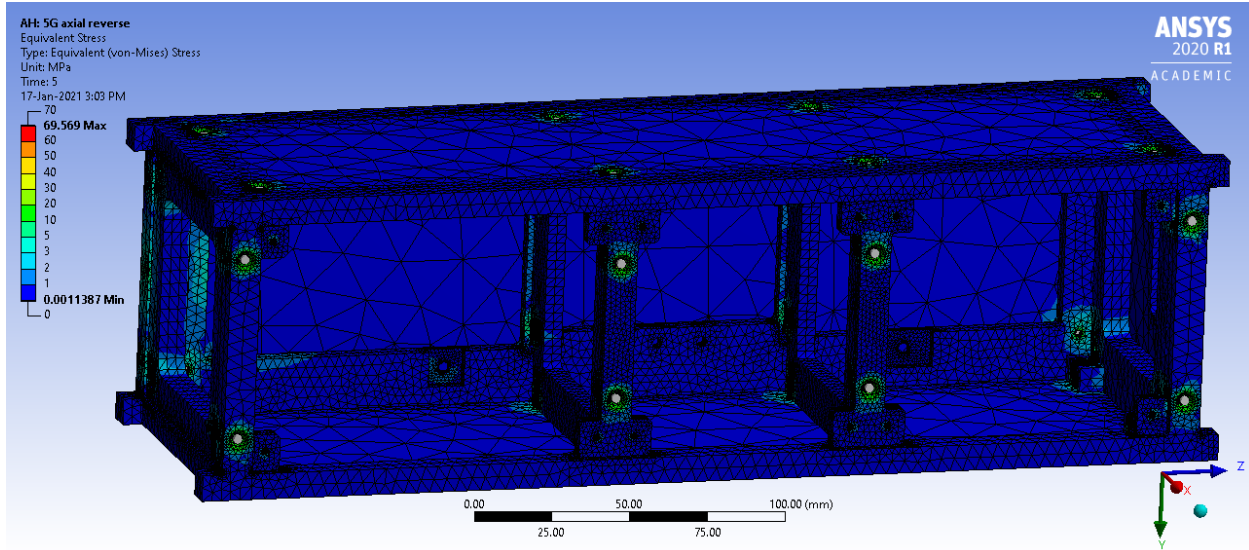


Figure 21: Phase 3 full model 5G +Z direction inertial load maximum stress

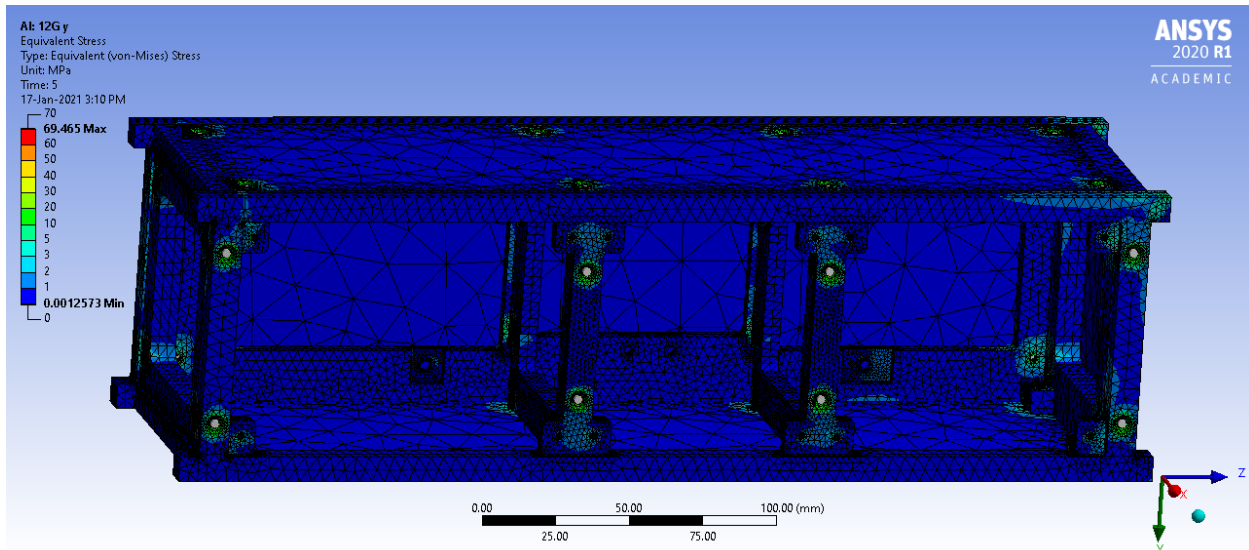


Figure 22: Phase 3 full model 12G Y direction inertial load maximum stress

The results of the completed inertial load analysis are summarized in Table 5. The maximum stress across all the load cases is due to the preload of the same screw in the +Y side cover panel. The maximum stress recorded in the PayBay screw is also negligible in each test case. Thus, the payload passes preliminary inertial load flightworthiness testing in all primary cases, and the Y axis secondary case.

Table 5: Phase 3 inertial load FEA results summary

LOAD DIRECTION	-Z	+Z	Y
ACCELERATION	15G	5G	12G
MAX STRESS (MPA)	69.6	69.6	69.6
MAX PAYBAY SCREW HOLE STRESS (MPA)	3.2	8.8	4.0

Modal analysis of the payload has been completed. Random vibration analysis is facing technical difficulties and requires further work. Modal analysis is set to search for the first 25 modes, starting at 0 Hz and ending at 4000 Hz. The modal analysis results are summarized in Table 6.

Table 6: Phase 3 modal analysis EMPF summary

Ratio of Effective Mass to Total Mass							
Mode	Frequency [Hz]	X Direction	Y Direction	Z Direction	Rotation X	Rotation Y	Rotation Z
1	718.36	4.6519e-002	5.2206e-005	2.1221e-004	1.7483e-006	9.8817e-004	1.5306e-003
2	833.26	8.2769e-008	8.3179e-002	2.9197e-005	6.0052e-003	2.1e-006	2.4345e-003
3	836.77	1.1713e-003	5.3348e-003	1.7189e-003	3.0078e-004	3.7391e-004	1.6153e-004
4	885.66	0.76722	2.7553e-004	5.6106e-002	3.5954e-005	0.21115	5.0235e-005
5	981.35	1.5871e-003	0.70445	4.2903e-003	0.14802	6.0236e-004	9.1032e-002
6	1025.1	4.2986e-002	4.4586e-003	0.90445	9.2256e-004	2.2898e-002	1.9574e-004
7	1189.8	4.7506e-003	1.1178e-003	1.7284e-004	9.596e-004	1.1011e-003	5.1862e-003
8	1265.6	4.2297e-005	3.3446e-004	6.3726e-004	4.9138e-003	4.5174e-005	4.3256e-003
9	1279.	1.5744e-005	7.1207e-004	3.5216e-004	1.4762e-002	4.8509e-007	1.5827e-002
10	1297.3	7.2059e-011	8.4932e-004	3.9791e-005	2.3033e-005	1.419e-006	6.164e-003
11	1309.5	3.9163e-006	6.0898e-004	4.2787e-005	6.3693e-005	1.3757e-005	5.0401e-003
12	1547.8	3.1084e-003	1.2697e-004	1.578e-004	8.285e-005	3.5047e-002	6.1222e-005
13	1592.9	1.9785e-004	1.3328e-003	8.6665e-005	1.5026e-003	9.7709e-003	1.09e-002
14	1634.9	5.6124e-005	2.8811e-002	5.2433e-005	6.1316e-002	8.2343e-004	0.56778
15	1709.9	4.7587e-007	1.618e-003	8.0625e-006	8.4359e-003	1.7722e-003	9.2866e-003
16	1735.5	9.6935e-004	1.7186e-004	2.9668e-003	1.5551e-003	1.7934e-003	1.2027e-003
17	1761.5	5.5056e-005	9.2562e-003	1.2207e-005	0.10948	8.6044e-004	3.5591e-002
18	1848.	1.1969e-002	2.9453e-007	3.6155e-003	2.1992e-004	0.41079	6.1355e-009
19	1904.4	7.8048e-005	2.3456e-004	2.396e-005	2.1951e-002	4.5152e-003	2.1704e-003
20	1968.6	1.2514e-005	6.4855e-003	2.7663e-005	0.18577	9.6036e-004	3.0846e-002
21	1992.7	1.9413e-007	3.6956e-003	6.6851e-006	2.4832e-002	2.1096e-004	1.9472e-002
22	2263.7	3.9508e-004	1.6002e-004	6.332e-005	2.1376e-005	3.5269e-005	3.7428e-004
23	2293.7	2.433e-004	1.1323e-005	6.15e-007	2.8592e-003	2.534e-005	7.7648e-004
24	2301.5	1.7203e-006	4.6907e-004	2.3428e-005	7.0444e-006	8.433e-004	1.3994e-003
25	2318.	1.3775e-005	6.2842e-003	2.1154e-006	1.1745e-002	6.3692e-005	2.7802e-004
Sum		0.8814	0.86003	0.9751	0.6058	0.70469	0.81209

The first mode is at 718.36 Hz. Mode 4 is the first significant mode with a frequency of 885.66 Hz and 77% X direction EMPF. Mode 4 shape is shown in Fig. 23. Mode 5 is the primary Y direction mode with a frequency of 981.35 Hz and 70% EMPF. Mode 5 shape is shown in Fig. 24. Mode 6 is the primary Z direction mode with a frequency of 1025.1 Hz and 90% EMPF. Mode 6 shape is shown in Fig. 25. All these modes are substantially greater than the 100 Hz minimum threshold typically required of CubeSats [12], [13].

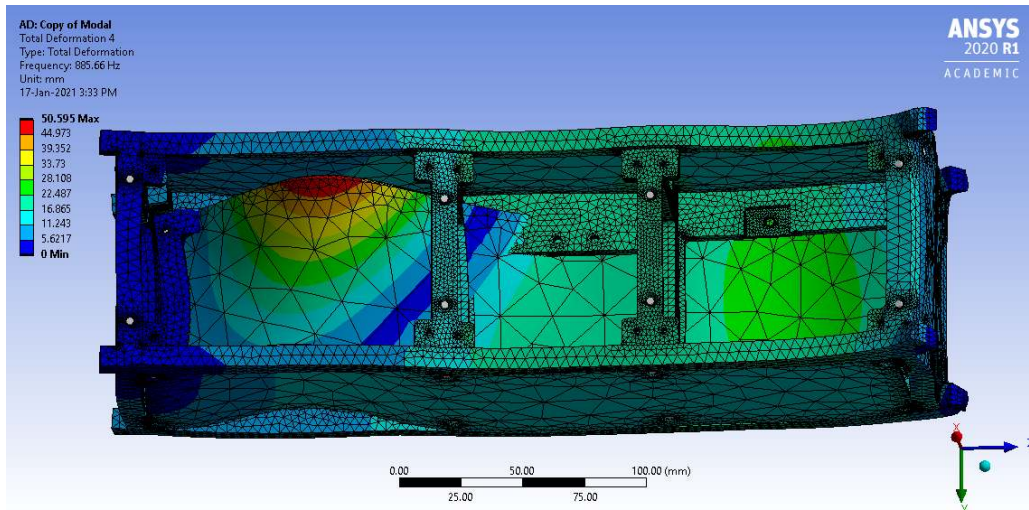


Figure 23: Phase 3, Mode 4 shape (885.66 Hz)

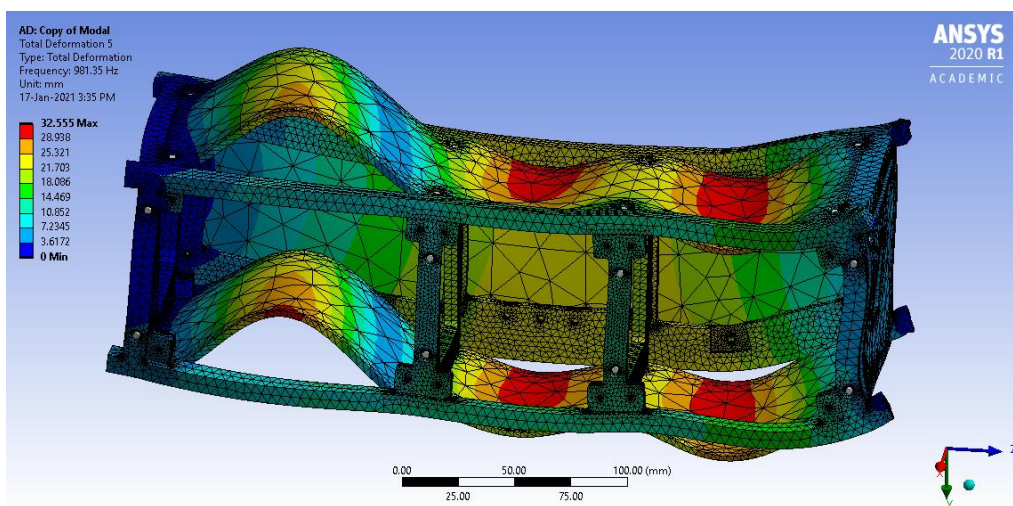


Figure 24: Phase 3, Mode 5 shape (981.35 Hz)

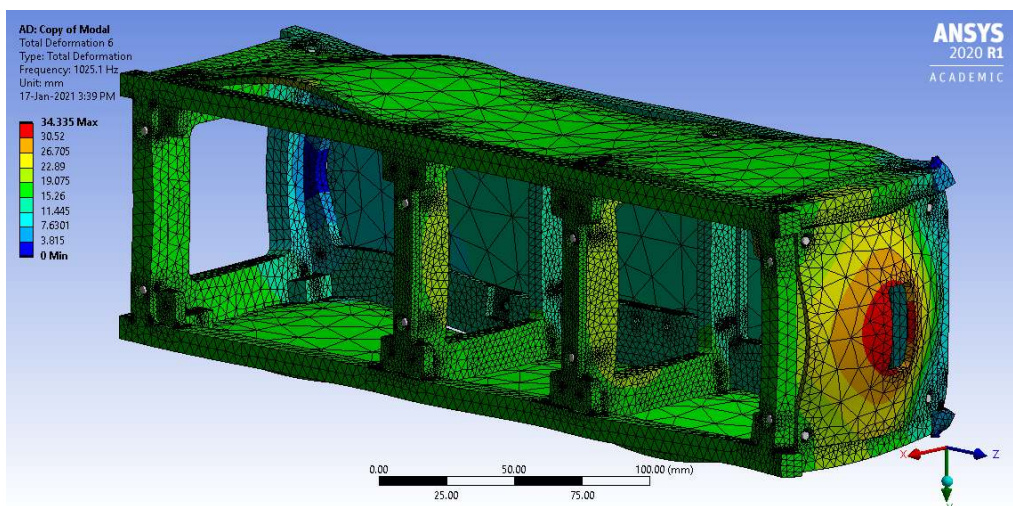


Figure 25: Phase 3, Mode 6 shape (1025.1 Hz)

The CEMPF of the X, Y, and Z directions are 88%, 86 %, and 98%, respectively. Each direction exceeds the 80% CEMPF rule-of-thumb for adequately characterizing the dynamic behaviour of a system. Mode 25 has a frequency of 2318 Hz, already exceeding the 2000 Hz upper bound of random vibration analysis; further modal analysis is superfluous.

Random vibration analysis is reportedly solved within Ansys. However, stress, fatigue life, and fatigue damage plots cannot be generated.

The following settings are to be used when the random vibration plots are generated. The maximum equivalent stress plot is set to 3- σ scale factor to display 99.73% of the stress expected. The payload is exposed to this vibration environment for 1 minute to determine fatigue damage in accordance with protoflight specifications in [7].

The payload passes preliminary dynamic flightworthiness as all relevant modes occur above 100 Hz. Further random vibration analysis is required to determine fatigue life of the CubeSat structure.

3.3 Remaining Challenges

Remaining challenges and work include running the last of the inertial load cases and completing the random vibration analysis. Inertial loading in +X and -X directions need to be run.

The problems with generating result plots in the random vibration analysis require further attention. Stress, fatigue damage, and fatigue life plots will be important in completing the preliminary flightworthiness verification of the payload. The error message given when attempting to generate these plots is “An error occurred inside the POST PROCESSING module while reading the result file”. Thus, it is suspected that the failure is due to user error. The material properties may lack the required information to determine the fatigue behaviour of the payload. The Granta Design Sample Materials library lacks fatigue information for the aluminium and steel alloy materials currently used in the model. Either fatigue data will need to be researched and added to the model, or mechanically similar materials with included fatigue data will need to replace the current material assignments.

4.0 Conclusions

Preliminary flightworthiness evaluation of the payload under inertial loading and modal analysis of the payload are completed using Ansys FEA. Internal modules are simplified to point masses and their behaviour is ignored in this analysis.

For inertial loading, the payload is subjected to three different test cases: 15G in -Z, 5G in +Z, and 12G in Y. In all test cases, the maximum stress does not exceed 70 MPa; no yielding occurs. Maximum stress occurs due to the screw preloads. Thus, the payload is deemed flightworthy for inertial load cases.

For modal analysis, there are four relevant modes: the first mode, and the primary X, Y, and Z axis modes. The first mode's frequency is 718.36 Hz and EMPF is negligible in all axis. The primary X axis mode's frequency is 885.66 Hz and EMPF is 77%. The primary Y axis mode's frequency is 981.35 Hz and EMPF is 70%. The primary Z axis mode's frequency is 1025.1 Hz and EMPF is 90%. All 4 relevant modes have frequencies above 100 Hz. Thus, the generalized dynamic behaviour of the payload is considered flightworthy.

As a result, no changes to the payload are taken.

This project prescribes a viable format for evaluating subsystem flightworthiness. Through a multistep approach, a simplified model is iteratively detailed until it sufficiently represents the subsystem in question. The intent is to minimize solver runtime by reducing the opportunities to add unnecessary complexity.

A process for specifying test specifications is detailed in this project. Using this process, flight and simulation data is analyzed to determine inertial and random vibration loads.

5.0 Next Steps

The next steps are to evaluate flightworthiness of the internal modules and final payload design. Following a similar process to that detailed in this report, the internal modules require FEA analysis to verify success under inertial and random vibration loads. Similarly, the finalized payload design will need to be subject to the same scrutiny as in this report. Ansys random vibration analysis is required to investigate the payload's fatigue behaviour. This entails resolving the challenges mentioned in section 3.3 of this report.

Physical vibration testing of the flight hardware should be conducted, if viable. This will require access to a shaker table capable of shaking a 4 kg system. Additional mounting hardware to couple the payload to the shaker table will need to be designed and manufactured. Aside from timing and access to testing equipment, testing viability is primarily derived from the risk associated with the internal modules; the cost and availability of replacement components may substantiate deadweight taking the place of flight hardware for random vibration testing.

References

- [1] Waterloo Rocketry. waterloorocketry.com (accessed Jan. 14, 2021).
- [2] K. Tam, “Design of a CubeSat Structure for Sounding Rocket Payloads,” Waterloo Rocketry Team, University of Waterloo, Ontario, Canada, 2020.
- [3] The CubeSat Program, California Polytechnic State University, California, USA. *CubeSat Design Specification*, rev 13. Accessed: Jan. 12, 2021 [Online]. Available: cubesat.org
- [4] C. Belle. “Mass Challenge for CubeSats.” Talk Satellite – Americas. www.talksatellite.com/Americas-A10085522.htm (accessed Jan. 15, 2021).
- [5] *Harris’ Shock and Vibration Handbook*, 5th ed., McGraw-Hill, 2002.
- [6] Ansys Learning. “Mode Participation Factor and Effective Mass — Lesson 4.” YouTube. www.youtube.com/watch?v=TO0gD-GpTfU (accessed Jan. 15, 2021).
- [7] *Payload Vibroacoustic Test Criteria*, NASA-STD-7001B, NASA, USA, 2017.
- [8] T. Sarafin, P. Doukas, M. Browning, L. Demchak. *Part 7: Designing a Small Satellite to Pass the Vibration Test*, rev. B (2020). Accessed: Jan. 7, 2021. [Online]. Available: www.instarengineering.com/pdf/Instar_Vibration_Testing_of_Small_Satellites_Part_7.pdf
- [9] *Product Verification Requirements for Launch, Upper Stage, and Space Vehicles*, MIL-STD-1540D, DoD, USA, 1999.
- [10] “VibrationData Toolbox Software.” enDAQ. endaq.com/pages/vibration-shock-analysis-software-vibrationdata-toolbox (accessed Jan. 10, 2021).
- [11] T. Sarafin, P. Doukas, M. Browning, L. Demchak. *Part 5: Random Vibration Testing*, rev. B (2017). Accessed: Jan. 7, 2021. [Online]. Available: www.instarengineering.com/pdf/Instar_Vibration_Testing_of_Small_Satellites_Part_5.pdf
- [12] N. Velenosi, M. Khor, M. Sabzalian, A. Elawad, N. Mahendrarajah, “MASS: A Modular Assembly Satellite Standard for 3U CubeSats,” Space Concordia, Concordia University, Montreal, Quebec, Canada, 2015.
- [13] A. Ampatzoglou, V. Kostopoulos, “Design, Analysis, Optimization, Manufacturing, and Testing of a 2U CubeSat,” *International Journal of Aerospace Engineering*, Volume 2018, doi: 10.1155/2018/9724263.
- [14] M. Khor, private communication, Jan. 2021.

- [15] A. Finklestein. "Understanding Mass Participation Factor Results in Frequency Studies." Computer Aided Technology. www.cati.com/blog/2019/11/understanding-mass-participation-factor-results-in-frequency-studies (accessed Jan. 12, 2021).
- [16] Ansys. ansys.com (accessed Jan. 15, 2021).
- [17] A. M. N. Elmekawy. (2018). Lecture 3 Meshing 2 [PDF]. Available: drahmednagib.com/CAD_2018/Lecture_3_Meshing_2.pdf
- [18] *Design, Test, & Evaluation Guide*, rev. 1, Launch Canada, Canada, 2021.
- [19] Spaceport America Cup, USA. *Intercollegiate Rocket Engineering Competition Rules & Requirements Document*, rev. E (2020). Accessed: Jan. 3, 2021 [Online]. Available: www.soundingrocket.org/uploads/9/0/6/4/9064598/sa_cup_irec_rules___requirements_document_20201006_rev_e.pdf
- [20] J. Robinson, "CAD Standard," Waterloo Rocketry Team, University of Waterloo, Ontario, Canada, 2019.
- [21] Tohnicchi America Corp., Buffalo Grove, Illinois, USA. *Bolt Tightening*, vol. 9. Accessed: Jan. 3, 2021. Available: www.tohnicchi.com/pdf/02-bolt-tightening.pdf
- [22] özgün. "Is Your Mesh Good Enough?" MecHead. www.mechead.com/mesh-good-enough/ (accessed Jan. 12, 2021).
- [23] Engineering ToolBox. "Friction and Friction Coefficients." www.engineeringtoolbox.com/friction-coefficients-d_778.html (accessed Jan. 1, 2021).
- [24] D. D. Fuller. *Coefficients of Friction*. Accessed: Jan. 1, 2021. [Online]. Available: web.mit.edu/8.13/8.13c/references-fall/aip/aip-handbook-section2d.pdf